

EXPLAINED EXAMPLE 1 – P-v-T EQUATION OF STATE

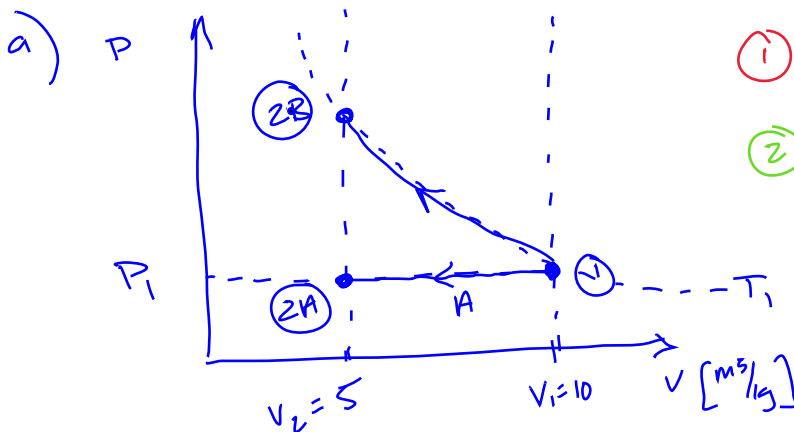
① An inventor is testing out different processes for cooling air in a piston-cylinder arrangement from the same initial state. In both processes, the mass-specific volume of the chamber reduces from $v_1=10 \text{ m}^3/\text{kg}$ to $v_2=5 \text{ m}^3/\text{kg}$, but Process A is isobaric and Process B is isothermal. Answer the following questions about these two processes if the temperature at state 1 is $T_1=1000 \text{ K}$. For air, $R=287 \text{ J/kg-K}$ and $c_v=714 \text{ J/kg-K}$. List your assumptions.

② a. Draw both processes on the same P-v diagram and label the state points – State 1, State 2A (after Process A), and State 2B (after process B).

b. Calculate T_2 and P_2 for both processes.

① Given : $v_1 = 10 \text{ m}^3/\text{kg}$
 $v_2 = 5 \text{ m}^3/\text{kg}$
 $P_{2A} = P_1$
 $T_{2B} = T_1$
 $T_1 = 1000 \text{ K}$
 $R = 287 \text{ J/kg-K}$
 $c_v = 714 \text{ J/kg-K}$

② Assume : air $\rightarrow$ ideal gas
 : quasi-steady
 : Process A $P_2 = P_1$
 : Process B $T_2 = T_1$
 : sealed, $M = \text{const}$



① Process A: $P = \text{const}$

② Process B: $T = \text{const}$

$$Pv = RT$$

$$P = \frac{RT}{v}, \quad P \propto \frac{1}{v} \text{ if } T = \text{const}$$

① b) Process A :

$$\text{State 1 : } P_1 v_1 = \frac{RT_1}{v_1} \rightarrow P_1 = \frac{RT_1}{v_1} = \frac{(287)(1000)}{10} \Rightarrow P_1 = 28,700 \text{ Pa}$$

$$\text{State 2 : } \boxed{P_2 = P_1 = 28,700 \text{ Pa}}$$

$$T_2 = \frac{P_2 v_2}{R} = \frac{(28700)(5)}{287} \Rightarrow \boxed{T_2 = 500 \text{ K}}$$

Probs B: $T_1 = 1000 \text{ K}$

②

$$P_1 = 28,700 \text{ Pa}$$

→ State 2: $T_2 = T_1 = 1000 \text{ K}$

$$P_2 = \frac{RT_2}{V_2} = \frac{(287)(1000)}{5} \Rightarrow P_2 = 57,400 \text{ Pa}$$

↑ $P_{2B} > P_{2A} \rightarrow \text{check graph}$